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Original Research

Optimising clinical information extraction: a comparative study of retrieval-augmented generation techniques in clinical notes

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ABSTRACT

Extracting clinically meaningful information from free-text notes in specific clinical settings, such as Australian aged care facilities, remains challenging due to the heterogeneity of text documents, which lack standardised structure and terminology. Retrieval-augmented generation (RAG) can improve the precision and grounding of large language model (LLM) outputs; however, the choice of retrieval strategies remains understudied despite its critical importance for clinical information extraction (IE). Using real-world clinical notes from Australian aged care facilities, we systematically compare six retrieval methods within a unified RAG pipeline: sparse retrieval (BM25), dense retrieval (bi-encoder embeddings), dense retrieval with cross-encoder rerank (abbreviated as dense reranking), dynamic linear fusion of sparse and dense scores, reciprocal rank fusion (RRF), and hybrid coarse-to-fine reranking. We evaluate these strategies on two clinical named entity recognition tasks, extracting agitation symptoms in dementia (n = 208) and identifying malnutrition risk factors (n = 208) across five dimensions: context relevance, answer quality, source faithfulness, contextual diversity, and item-level accuracy. A repeated-measures ANOVA reveals that reranking and hybrid ensemble strategies significantly outperform both standalone sparse (BM25) and dense retrieval across both tasks. For agitation extraction, Dense reranking achieves the highest Answer F1 (0.946), Context Diversity (0.895) and Item-level Accuracy (0.963). For malnutrition, ensemble methods yield the best Item-level Accuracy (0.944), followed closely by dense reranking. Through error analysis, we identify three LLM error types in clinical named entity recognition: intrinsic hallucination, extrinsic hallucination, and false negatives, and elucidate how RAG mitigates each. These findings demonstrate that reranking-based retrieval substantially enhances the performance of RAG pipelines for clinical information extraction. It offers a practical approach for improving automated analysis of unstructured clinical text. Our four-stage experimental workflow – document indexing, context retrieval, LLM generation, and structured output formatting – provides a replicable framework for future clinical information extraction research and downstream predictive modelling.

1. Introduction

Due to a lack of standardised terminology and structure in nursing documentation in Australian residential aged care settings, the major clinical information is locked in free-text clinical notes. These notes are highly heterogeneous, characterised by varying writing styles and languages across staff [1], inconsistent terminology and structure across facilities [2], frequent abbreviations, context-dependent acronyms, and typographical errors. Extracting essential clinical variables from such unstructured text is therefore a significant challenge, one that clinical named entity recognition (NER) methods in natural language processing are well positioned to address [3].

Large language models (LLMs) offer powerful capabilities for extracting structured information from unstructured clinical notes. However, foundation LLMs that lack domain-specific pretraining can produce inaccurate, hallucinated, or biased outputs [4], raising concerns about their deployment in high-stakes healthcare contexts, where precision, contextual understanding, and domain-specific knowledge are paramount [5]. Retrieval-augmented generation (RAG) addresses these limitations by incorporating external evidence into the generation process, enabling more accurate and contextually grounded clinical information extraction [5,6]. Importantly, Vithanate et al. systematically

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compared training methods for adapting generative LLMs to clinical named-entity recognition in aged-care nursing notes from the same dataset used in the present study [5,6]. Their findings demonstrated that few-shot learning with RAG achieved performance comparable to zero-shot learning with parameter-efficient fine-tuning (PEFT), despite the two approaches relying on fundamentally different adaptation mechanisms, external evidence retrieval versus internal parameter modification. This equivalence suggests that, for the clinical information extraction (IE) tasks examined here, the retrieval component of a RAG pipeline may offer as much leverage for performance improvement as domain-adapted parameter tuning, without the computational overhead or technical expertise associated with fine-tuning.

While that study established the viability of RAG as an adaptation strategy, it did not examine how different retrieval strategies within the RAG pipeline affect extraction quality. This gap motivates the present investigation: which retrieval configuration best supports contextual grounding and factual faithfulness in clinical IE?

The RAG framework integrates two core processes, retrieval and generation [7]. During retrieval, a knowledge source, which may consist of external documents, clinical guidelines, or clinical records [6], is precoded by a document encoder into dense vector representations. These vector embeddings are stored in a vector database for efficient similarity-based retrieval [7,8]. When a user submits a query, the retrieval module identifies the top-k most relevant documents through vector similarity matching, providing the LLM with dynamically accessed, pertinent context [9]. In the subsequent generation stage, both the original query and the retrieved documents are passed to a pre-trained decoder model such as GPT or Llama [6]. This model uses the retrieved content as external context to produce more relevant, factually grounded responses. This integration of queries and evidence is essential to enhancing the accuracy and safety of LLM applications in healthcare, where errors can have serious consequences.

Despite advances in both retrieval and generation technologies, the effectiveness of RAG pipelines remains fundamentally determined by the retrieval quality [10,11]. The relevance and specificity of the retrieved documents directly influence the accuracy of downstream outputs; irrelevant or low-quality retrievals can lead to erroneous, incomplete, or misleading generated content. Consequently, selecting and optimising the retrieval strategy is critical for producing clinically meaningful and trustworthy LLM outputs aligned with user intent. Various retrieval approaches have been developed, ranging from sparse lexical matching (e.g., BM25) to dense neural embeddings and hybrid ensemble methods. However, their comparative effectiveness for clinical information extraction tasks, particularly in specialised settings such as Australian residential aged care, remains understudied.

To address this gap, we systematically evaluate six retrieval strategies within a unified RAG pipeline for clinical named entity recognition using real-world clinical notes from Australian residential aged care facilities. We assess performance across two clinically significant tasks: extracting agitation symptoms in dementia and identifying malnutrition risk factors, and evaluate five dimensions: context relevance, answer quality, source faithfulness, contextual diversity, and item-level accuracy. Beyond identifying optimal retrieval strategies for clinical RAG applications, this study contributes to knowledge about the LLM error types and a replicable four-stage experimental workflow (document indexing, context retrieval, LLM generation, and structured output formatting) that can be extended to future clinical information extraction research and downstream predictive modelling.

2. Related work

2.1. Assessing linguistic diversity of clinical notes

Clinical notes express the same clinical concept through many different surface forms, directly affecting RAC retrieval performance. However, surface vocabulary breadth may overestimate the true diversity of clinical content if the semantic layer is not independently assessed. Sohn et al. [12] demonstrated that while the lexicon of clinical notes shows considerable variability across institutions, the overall semantics remains highly similar at the conceptual level.

Therefore, assessing linguistic diversity requires capturing lexical and semantic dimensions independently. Jarvis [13] identifies variability (the range of different word types) and disparity (the degree to which these word types differ semantically) as separable dimensions of text diversity, each requiring independent measurement [14]. Following this framework, we assess the linguistic diversity of clinical notes using two complementary measures: lexical diversity and intra-content semantic dispersion.

First, lexical diversity is computed as the type–token ratio, i.e., the ratio of unique word types to total tokens, to capture surface-level vocabulary variation [15]. Let $R = \{r_1, r_2, \ldots, r_k\}$ denote the set of text segments in a clinical corpus, $V(R)$ the set of unique tokens after tokenisation, and $T(R)$ the total token count. Lexical diversity (LD) is defined as:

$$Lexical\ \ Diversity(LD) = \frac{|V(R)|}{T(R)}$$

Second, intra-content semantic dispersion is computed as the mean pairwise cosine distance between document-level term-frequency vectors, capturing whether documents occupy distinct regions of the distributional semantic space or cluster around similar content.

Concordance between the two measures indicates that vocabulary breadth reflects genuine content heterogeneity, whereas discordance suggests that substantial lexical variation conceals largely repetitive concepts. This diagnostic assessment provides essential context for interpreting the retrieval comparisons reported in subsequent sections.

2.2. Retrieval strategies in RAG pipeline

Prior work has demonstrated the potential of LLMs and RAG for extracting biomedical knowledge and clinical signals from EHRs [16]; however, direct comparisons of these retrieval strategies for clinical IE in specific clinical settings, such as aged care, remain rare. Therefore, this section reviews six retrieval strategies, tracing a path from purely lexical matching to interaction-heavy reranking and, finally, to hybrid strategies. The six strategies include (R1) sparse lexical retrieval (BM25), (R2) dense bi-encoder retrieval, (R3) two-stage dense reranking with a cross-encoder, (R4) dynamic linear fusion of sparse and dense scores, (R5) reciprocal rank fusion (RRF), and (R6) hybrid BM25 to dense rerank (coarse-to-fine).

R1: Sparse Retrieval (BM25)

Sparse retrieval (BM25) is a classical retrieval method. It ranks documents by term frequency-inverse document frequency TF-IDF ranking and probabilistic term weighting [17]. It uses lexical matching to identify relevant documents, assuming that the frequency and distribution of exact terms can reflect relevance. This method uses the BM25 API to score each document against the tokenised query using two tunable parameters: **k** for term frequency saturation and **b** for document length normalisation. The top-3 highest-scoring documents are concatenated to form the context, which is then passed to an LLM for generation. In domains such as structured clinical narratives, where terminology is stable, and cues recur in consistent lexical forms, this method works well. It is interpretable, fast, transparent, and easy to scale [16,17]. However, clinical text can be messy, containing abbreviations, paraphrases, and local shorthand. These deviations from standard terminology can undermine BM25′s reliance on surface matching [18].

R2: Dense Retrieval Using Semantic Embeddings

Dense retrieval applies neural embeddings to capture semantic similarity between queries and documents in a high-dimensional vector space [18]. This approach encodes queries and documents with Sentence-BERT (e.g., the all-mpnet-base-v2 model) into 768-dimensional vectors [19] and retrieves the top-3 document snippets by cosine similarity. Dense retrieval is particularly suited to mining clinical notes, where concepts may be expressed in synonyms, paraphrases, or shorthand. The model can align meaning even when the surface lexical form differs [19–21]. The trade-off is less interpretable; performance can degrade without pretraining tailored to biomedical or clinical language [22,23], and extensive document collections demand careful engineering (e.g., approximate nearest-neighbour indexing) to manage storage and latency [24]. Still, when lexical signals are noisy, dense methods often uncover what BM25 misses.

R3: Dense Retrieval with Cross-encoder Reranking (Dense Reranking)

Dense retrieval with cross-encoder reranking excels at capturing broad semantic relationships by independently encoding the query and document, enabling flexible matching across diverse contexts [25,26]. It adopts a coarse-to-fine strategy that balances retrieval breadth with ranking depth. A Sentence Transformer bi-encoder first proposes the top-10 candidates by cosine similarity; a transformer cross-encoder then reranks these candidates, and the top-3 proceed to the LLM for generation [27,28]. The final cross-encoder reranking step incorporates query-document interactions, yielding context-sensitive precision gains over bi-encoder retrieval [26,28]. The result is a balanced, broad semantic recall, followed by precise, context-aware scoring. It is thus well-suited to healthcare, where small textual cues can flip clinical meaning [29]. Its main limitation is computational cost. Cross-encoder inference is slower and more challenging to scale than bi-encoder models, demanding batching, parallelisation, or distillation to meet real-time constraints [30].

R4: Dynamic Linear Fusion of Sparse and Dense Retrieval

Rather than choosing between lexical and semantic signals, dynamic linear fusion retrieval lets the query decide which to use. It linearly fuses BM25 scores with dense cosine similarities after per-query min–max normalisation. It uses the dispersion of each score distribution (e.g., the standard deviation) as a proxy for its discriminative power on that query and weights the fusion accordingly; the top-3 fused results are passed to the LLM [28,31]. This adaptive scheme often stabilises performance across heterogeneous texts. If the query is lexically specific, BM25 is appropriate; when it is semantically broad or ambiguous, dense similarity performs better. The limitation is its reliance on careful calibration. If both inputs are noisy, the combination can amplify errors, and without explicit token-level interactions, it may still miss subtle semantic nuances [26,32].

R5: Reciprocal Rank Fusion (RRF)

Reciprocal Rank Fusion (RRF) is a robust and straightforward method that combines search results from different systems by ranking documents based on their positions rather than their scores [22]. In the implementation by Luan et al. [22], RRF combines BM25 and dense retrieval outputs using SentenceTransformer embeddings. The top-3 ranked documents are then passed to the LLM [22]. By avoiding complex score normalisation, RRF performs well across systems with different scoring methods and achieves strong performance in both general and biomedical retrieval tasks [28,33]. However, RRF only considers rank positions, which can overvalue marginally relevant documents. It also lacks query-specific adaptation, limiting its effectiveness for complex clinical queries [22,28].

R6: Hybrid BM25 to Dense Bi-encoder Rerank

Finally, the Hybrid BM25-to-dense bi-encoder rerank uses a pragmatic, cascaded retrieval architecture. First, BM25 retrieves a focused candidate pool (top-10) by exploiting exact term matches and document length normalisation; a dense bi-encoder then reranks these candidates via cosine similarity, and the top-3 reach the LLM [21,30,33,34]. This design combines BM25′s keyword sensitivity with the semantic flexibility of embeddings while keeping the computer in check by scoring densely only within a small pool [10,22]. The weakness is an early-exit failure mode: if BM25 misses a relevant document, the dense stage cannot recover it [33]. Because the dense rerank is a bi-encoder rather than a cross-encoder, some fine-grained interactions remain out of reach [27].

A subtle trade-off is unavoidable in selecting retrieval strategies. If a consistent pattern is available in the text, lexical methods are optimal for their efficiency and precision. Semantic methods are resilient to variation in language expression, whereas interaction models add nuance at a computational cost, and fusion strategies can balance strengths across methods but rely on careful calibration. What is still missing in the clinical IE literature is a direct, comparative evaluation of these strategies on the same tasks and corpora. This gap motivates our study: to test which retrieval strategy most reliably uncovers the nuanced, context-specific evidence needed to extract clinical information from unstructured clinical notes. The answer to the question has direct implications for the application of LLM in healthcare. It shapes how quickly and accurately we can help clinicians translate clinical information captured in free-text clinical notes into actionable information for real-world care decision-making.

2.3. Evaluation methods

We designed five complementary evaluation metrics to assess the performance of the six retrieval strategies: Context F1, Answer F1, Faithfulness F1, Contextual Diversity, and Item-level Accuracy. Each metric addresses a distinct dimension of retrieval system performance.

E1: Context F1

Context F1 tests the semantic similarity between the retrieved text and the annotated ground-truth text. It uses BERTScore, which embeds both texts into contextualised vector representations and then computes precision, recall, and F1 based on token-level semantic similarity [35]. BERTScore's ability to recognise paraphrasing and semantic similarity makes it suitable for analysing free texts, such as clinical notes [36]. A high Context F1 score reflects a strong match between the retrieved content and the reference evidence, indicating that the retrieval strategy successfully identifies relevant and comprehensive information.

E2: Answer F1

Answer F1 evaluates the semantic correctness of the LLM-generated answer by comparing it with the gold-standard annotation using BERTScore [35], which measures token-level similarity. In this study, BERTScore was computed using a general-purpose pretrained BERT model from the original BERTScore implementation. This model is distinct from the specialised embedding models used in the retrieval stage.

BERTScore precision measures how closely the generated content aligns with the gold-standard answer. Recall captures the proportion of the gold-standard content included, and Answer F1 combines precision and recall to assess overall semantic fidelity. This makes Answer F1 an indicator of the quality of clinical information generated by an LLM, ensuring alignment with expert-labelled data [37,38]. Unlike lexical metrics such as ROUGE [39], Answer F1 is robust to frequent paraphrasing and linguistic variation in unstructured clinical notes, making it better suited to evaluating semantic accuracy in clinical text generation.

E3: Faithfulness F1

Faithfulness F1 measures the degree to which the generated text is grounded in the retrieved evidence. It uses BERTScore to compare the semantics of the generated text against the retrieved text (see Formula 1) [35]. When a model scores highly on faithfulness, its answers are backed by solid evidence, whereas low scores suggest the model may have generated unsupported or misleading content [40,41]. In clinical and biomedical NLP, maintaining faithfulness is critical, as unfaithful outputs can distort patient data or misinterpret health risks [42]. Faithfulness F1 aligns with recent advances in factual consistency evaluation, including QA-based metrics such as QAFactEval [43], comprehensive surveys such as TRUE [44], and early model-based approaches to verify factual consistency in summarisation [45].

To formally define the computation of Faithfulness F1, we measure the degree to which the LLM-generated answer is supported by the retrieved context using BERTScore. Let a denote the generated answer and c denote the retrieved context. BERTScore yields precision $P_f(a,c)$ and recall $R_f(a,c)$, and faithfulness is summarised using the harmonic mean:

$$F_1^{faith}(a,c) = \frac{2P_f(a,c)R_f(a,c)}{P_f(a,c)+R_f(a,c)}$$

E4: Contextual Diversity

Contextual diversity extends the corpus-level characterisation in Section 2.1 to the retrieval output, quantifying the degree of lexical variability across the top-k retrieved segments for each instance. Higher contextual diversity indicates that the retrieval system is capturing evidence from a broader terminological space rather than returning redundant content, thereby increasing the likelihood of surfacing complementary cues distributed across fragmented clinical notes.

For statistical comparison across retrieval methods, only lexical diversity is used, defined identically to Section 2.1 but computed over retrieved segments rather than the source corpus:

$$Contextual\ \ Diversity(CD) = \frac{|V(R)|}{T(R)}$$

Where R denotes the retrieved segments for a given query. The dual-measure triangulation strategy employed in Section 2.1 is not repeated here, as the analytical question differs. Corpus characterisation required both measures because the relationship between lexical and semantic layers was unknown *a priori,* and as Sohn et al. (2014) showed [12], the two can diverge in clinical text. Once that relationship has been established for a given corpus, it serves as a fixed baseline. At the retrieval level, the question shifts to whether a given method selects terminologically varied segments from that material. Lexical diversity measures it adequately. All measures were computed at the instance level and summarised descriptively across the text set [14,46].

E5: Item-Level Accuracy

Item-Level Accuracy evaluates the correctly identified discrete clinical factors (e.g., risk factors) in the generated output. Mirroring granular evaluation practices in fine-grained IE tasks, this metric decomposes multi-faceted answers into atomic items and measures each item's semantic correspondence to the annotated items. Various researchers have implemented item-level accuracy. For example, Léchelle et al. (2018) introduced WiRe57, a benchmark for ensuring the precise extraction of individual tuples in open IE systems [47]. Wu et al. (2024) proposed an entity-centric evaluation framework, the AESOP metric, to compute entity-level set overlap [48].

To evaluate correctness at the level of normalised entity items, we compute item-level accuracy based on semantic matching between predicated items and ground-truth items. Let $\widehat{Y} = \{\widehat{y}_1, \ldots, y_n\}$ be the set of predicted items obtained by normalising and splitting the generated answer, and let $Y = \{y_1, \ldots, y_n\}$ be the ground-truth item set. We embed each item using a sentence embedding model e (.) and compute cosine similarity:

$$sim\left(\widehat{y}_i, y_j\right) = \frac{e(\widehat{y}_i)\cdot e\left(y_j\right)}{\|e(\widehat{y}_i)\|\,\|e\left(y_j\right)\|}$$

A ground-truth item y_j is considered matched if there exists at least one predicted item $\widehat{y}_i$ such that $\max_i \text{sim}(\widehat{y}_i, y_i) \geq \tau$, where τ is a predefined threshold (set to 0.7 in this study). This matching process is fully automated and does not involve manual adjudication, ensuring reproducibility and consistency across all retrieval methods. Let

$$M = \sum_{j=1}^{n} 1\left[maxsim_i\left(\widehat{y}_i y_j\right) \geq \tau\right]$$

Denote the number of matched ground-truth items. Item-level accuracy is then defined as:

$$Item - Level\ \ Accuracy = \frac{M}{|Y|}$$

This metric is computed per instance and averaged across instances.

3. Methods

This study systematically evaluates six retrieval strategies within a retrieval-augmented generation (RAG) pipeline for an LLM to extract clinical information from clinical notes. The information retrieval task is named-entity recognition for two health conditions that older people in residential aged care facilities commonly experience: (1) symptoms of agitation in dementia, and (2) malnutrition risk factors. We rigorously evaluated the retrieval strategies in the RAG framework across five performance dimensions.

3.1. Ethics considerations

This study was approved by the University of Wollongong Human Research Ethics Committee (Reference No.: 2019/159). The de-identified clinical dataset was nursing progress notes gathered from 40 residential aged care facilities in New South Wales, Australia, between 2019 and 2021.

The research process consists of two phases: data preparation, a four-stage RAG-LLM pipeline, and model evaluation.

3.2. Phase 1: Dataset Preparation

The annotated dataset was derived from the previous studies conducted by the research group. The data was collected from 40 residential aged care facilities (equivalent to nursing homes in the US) belonging to a management group in New South Wales, Australia, in 2019. The Gold-standard test dataset for agitation in dementia was the dataset validated in [5,6], and the malnutrition risk factors dataset was based on [7]. Each Gold-standard test dataset contains 208 records.

3.2.1. Instance-level Heterogeneity Analysis

Following the method presented in Section 2.1, we computed heterogeneity measures at the instance level across the 208 evaluation records for each clinical task and reported summary statistics. First, lexical diversity was measured using the type-token ratio (TTR). Second, semantic dispersion was used to estimate the degree of semantic variability within each textual instance. For each record, the text was segmented into its constituent units, represented using term-frequency vectors, and pairwise cosine distances were calculated between segments. The mean upper-triangular cosine distance was then used to

represent intra-instance semantic dispersion, with higher values indicating greater semantic variation across segments. All heterogeneity measurements were calculated at the instance level across the 208 evaluation records in each clinical task, and summary statistics were then reported across the test set. To contextualise the heterogeneity observed in the clinical material, the same analysis pipeline was additionally applied to a normal-text reference corpus sampled from WikiText-103, a widely used general-domain English dataset derived from verified Good and Featured Wikipedia articles [49]. The normal-text corpus was sampled to match the evaluation size of the clinical dataset (n = 208), and identical preprocessing procedures and metric definitions were applied to ensure comparability between the clinical and normal-text corpora.

3.3. Phase 2: Four-stage RAG-LLM Pipeline

We implement a four-step RAG-LLM pipeline: Document ingestion and indexing (indexing), context retrieval and optimisation (retrieval), knowledge-grounded generation (generation), and structured output formatting (output), (see Fig. 1).

3.3.1. Document Ingestion and Indexing (Indexing)

The first task was to transform unstructured clinical notes into machine-readable knowledge representations (RAG files) that facilitated retrieval. The first step involved text cleaning and document segmentation into semantically coherent text segments. The second step was embedding generation using the Sentence-BERT (all-mpnet-base-v2) vector embedding model. The third step involved indexing these embeddings in a FAISS vector database [50] using a flat inner-product index (IndexFlatIP) to enable efficient nearest-neighbour retrieval by cosine similarity. When segmentation was required, source documents were partitioned into fixed-length segments of 250 tokens, a setting chosen to balance retrieval granularity, within-segment semantic coherence, and computational efficiency [16]. When segmentation was not required (e.g., shorter or structurally concise notes), the full document was treated as a single segment. The selected segments were then concatenated in descending order of retrieval rank to form the model input context.

3.3.2. Context Retrieval and Optimisation (Retrieval Optimisation)

To evaluate the impact of of the six retrieval strategies on clinical IE: (R1) sparse retrieval, (R2) dense retrieval based on semantic embeddings, (R3) dense retrieval + cross-encoder rerank (dense reranking), (R4) dynamic ensemble of sparse and dense retrieval, (R5) reciprocal rank fusion (RRF), and (R6) hybrid BM25 filter with dense bi-encoder rerank [30]. This comprehensive selection enabled a nuanced analysis of retrieval effectiveness across the two clinical tasks.

We used LlamaIndex 0.12.6 to orchestrate retrieval. For each query, we retrieved the top 3 ranked items using the strategy-specific scoring function. BM25 used sparse retrieval. Dense retrieval used cosine similarity between unit-normalised embeddings. Cross-encoder relevance logits were used for rerankers. Finally, fused or rank-aggregated scores were applied in hybrid methods [22]. Ties were resolved deterministically by prioritising lower document IDs [51].

Context retrieval began by embedding each query and then performing a similarity search using each of the six retrieval strategies above. This step grounded the LLM in accurate, contextually relevant text rather than relying solely on its parametric memory.

3.3.3. Knowledge-grounded LLM Generation (LLM Generation)

We implemented Llama3.1.8 B to generate responses, which integrated the retrieved text from the previous step, concatenated it with the query, and then injected the result into the input prompt. The model configuration includes nucleus sampling (top-p = 0.9) at a temperature of 0.1 and a repetition penalty of 1.2; other decoding limits (e.g., max new tokens and stop sequences) followed the task-specific output schema.

We adopted three-shot prompting as the primary strategy in our experiments, inspired by the work of Vithanage et **al.** [5]. Supplementary Table S1 lists the prompts used in this research.

3.3.4. Formatting the Structured Output (Output Formatting)

This stage was focused on transforming the LLM's free-form output into a structured format suitable for downstream tasks. Post-processing was conducted to parse the output into a standardised JSON schema, with distinct fields capturing (1) symptoms of agitation in dementia, and (2) malnutrition risk factors.

Given the heterogeneity of aged care clinical notes, we adopted a prompt-based schema specification approach rather than rigid, constrained decoding, as the latter assumes a degree of lexical and structural regularity in the source text that does not hold in our study context. Conversely, prompt-based schema specification allows the model to map semantically diverse inputs onto predefined clinical categories without imposing hard constraints that may fail when encountering unanticipated phrases. We acknowledge that constrained decoding remains a promising approach for improving consistency and can be implemented in future work once aged care documentation practices become more standardised.

3.4. Phase 3: Model Evaluation

3.4.1. Comparative Performance of RAG-augmented and Standalone LLM Approaches to Clinical Named Entity Recognition

To establish whether retrieval augmentation yields a measurable performance gain, we evaluated the LLM on the schema-driven extraction task using prompting alone, without any retrieved context. This no-retrieval baseline served as an ablation baseline, with retrieval as the sole experimental condition, using the same LLM, prompt template, decoding configuration, output schema, and evaluation procedure. Under this controlled design, observed performance differences between the baseline and retrieval-augmented conditions can be attributed to the contribution of retrieved evidence rather than to confounding variation

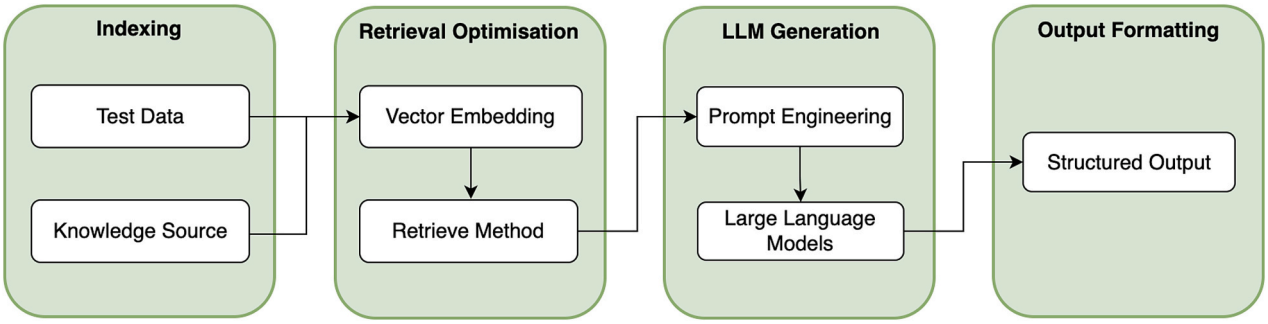


Fig. 1. Four-stage RAG-LLM pipeline and knowledge base construction.

in other pipeline components.

3.4.2. Evaluate Across Five Complementary Evaluation Dimensions

We evaluated system performance across five complementary dimensions for retrieval effectiveness, averaging per-instance scores over 208 test cases for each task. The unit of analysis is each nursing record. The micro-average score was implemented. For precision, recall and F1 score. We used a RAGAS-inspired rubric [23] to measure context F1, which involves computing precision, recall, and F1 based on the alignment between retrieved segments and annotated supporting evidence. Faithfulness F1 evaluates whether the generated answers are grounded in retrieved evidence, penalising contradictions and unsupported claims. Contextual diversity was measured using a lexical diversity measure computed over the top-k retrieved segments after z-normalisation within each task to ensure comparability. Finally, we used item-level accuracy to measure the proportion of gold-standard factors correctly identified in each instance, providing a fine-grained assessment of correctness.

3.5. Statistical analysis

We used an instance-level repeated-measures (RM) ANOVA [52] to compare performance differences among retrieval methods. Each test case was treated as a subject, and the six retrieval methods were treated as conditions within that subject to measure the five key metrics: Context F1 score, Answer F1 score, Faithfulness F1 score, Contextual Diversity, and Item-level Accuracy. After the primary analysis, we conducted post hoc pairwise comparisons using two-sided paired t-tests, adjusting for multiple comparisons with Holm's correction [53].

We summarised the results using compact letter displays (CLD), where bars with the same letter denote no significant difference within the panel [54]. We checked the score distributions to ensure the normality assumption was met before conducting an RM-ANOVA on task-level average scores from 208 test instances per task-method combination and 416 instances aggregated across both clinical tasks [5,29]. This methodological choice was consistent with prior evaluation frameworks in clinical IE. Pairwise post-hoc comparisons used two-sided paired t-tests with Holm adjustment ($\alpha = 0.05$); compact letter displays (CLD) above bars denote significance groups, where identical letters denote groups that are not significantly different within the group. In addition to statistical comparison of retrieval performance, we also conducted a heterogeneity analysis of the evaluation dataset and retrieved RAG contexts to characterise the linguistic and semantic variability present in the clinical narratives.

3.6. Experiment configuration

The experiment was conducted on the NCI Australia (National Computational Infrastructure Australia) Gadi supercomputing cluster [25,29], utilising nodes equipped with four NVIDIA Tesla V100 GPUs (32 GB of memory), 48 CPU cores, and 382 GB of RAM. The operating system and frameworks include Linux 4.18, PyTorch 2.1, Python 3.11.0, and LangChain 0.3.21.

4. Results

4.1. Heterogeneity comparison between clinical and normal text

The clinical notes exhibited significantly larger linguistic heterogeneity than the normal-text corpus. Clinical notes demonstrated higher lexical diversity (mean TTR = 0.58) than the normal-text corpus (mean TTR = 0.41, $p < 0.001$). Similarly, semantic dispersion within textual instances was greater for clinical notes (mean cosine distance = 0.73) than for the normal-text corpus (mean = 0.49, $p < 0.001$). These test results suggest that aged care clinical notes are significantly more heterogeneous than general domain text.

For agitation in dementia, retrieved contexts exhibited high lexical and semantic variability, with a mean lexical type–token ratio of approximately 0.71 and an average intra-context semantic dispersion of 0.82, reflecting substantial variation in language expression and contextual focus across nursing records. For malnutrition risk factor extraction, the retrieved contexts demonstrated moderate but non-trivial heterogeneity, with a mean lexical diversity of around 0.45 and a semantic dispersion of around 0.54. Across both tasks, each retrieved context comprised, on average, three semantically distinct segments, indicating that RAG inputs typically aggregated evidence from multiple narrative fragments rather than a single homogeneous passage.

4.2. Comparative performance of RAG-augmented and standalone LLM approaches to clinical named entity recognition

Without RAG, the LLM achieved Answer F1 scores of 0.702 for agitation symptom extraction and 0.722 for malnutrition risk factor extraction. The best-performing retrieval-augmented configuration (dense reranking) substantially outperformed this baseline, achieving F1 scores of 0.946 and 0.953, respectively ($p < 0.001$), an absolute gain of approximately 0.24 F1 points across both tasks (see Section 4.2). This performance gap demonstrates that RAG-augmentation is not merely beneficial but necessary for reliable schema-driven clinical information extraction from heterogeneous aged care nursing notes, where relevant evidence is sparse and unevenly distributed across free-text records.

4.3. Context F1

Statistically significant differences were found between retrieval methods for Context F1 in each clinical IE task and their combination ($p < 0.001$). For clinical IE in agitation in dementia (Fig. 2), dense-rerank and RRF achieved the highest scores (0.922 and 0.924), forming the top significance group. At the same time, Interaction-rerank and Ensemble had the lowest scores (0.824 and 0.832, respectively). For the malnutrition task, dense-rerank and RRF again performed best (0.896 and 0.894), whereas BM25 and Interaction-rerank were lowest (0.841 and 0.843). In the pooled analysis, dense-rerank and RRF remained the top groups, with no significant difference between them (0.909 and 0.910), while ensemble and interaction-rerank were significantly lower (0.839 and 0.833). These findings demonstrate that reranking-based retrieval provides more substantial contextual alignment than sparse or baseline dense retrieval.

4.4. Answer F1

Significant differences were also observed for Answer F1 ($p < 0.001$). In the agitation task (Fig. 3), dense-rerank achieved the highest Answer F1 (0.946), followed by RRF (0.935), while BM25 was lowest (0.881). For malnutrition, dense-rerank and RRF both formed the top group (0.953 and 0.944), with BM25 again at the bottom (0.871). The pooled analysis confirmed these trends, with dense-rerank and RRF leading (0.953 and 0.947) and BM25 lowest (0.838). These results highlight that high-quality retrieval directly enhances downstream answer precision and recall.

4.5. Faithfulness F1

Method differences were significant for Faithfulness F1 across tasks ($p < 0.001$). For agitation (Fig. 4), RRF and dense-rerank were statistically indistinguishable at the top (0.905 and 0.904), whereas interaction-rerank had the lowest score (0.804). For malnutrition, dense-rerank (0.858) achieved the best performance, followed by RRF (0.845), while BM25 and ensemble had significantly lower performance (0.787 and 0.789, respectively). The pooled analysis of two tasks yielded similar results, with dense-rerank achieving the highest (0.882) and interaction-rerank achieving the lowest (0.798). These findings confirm

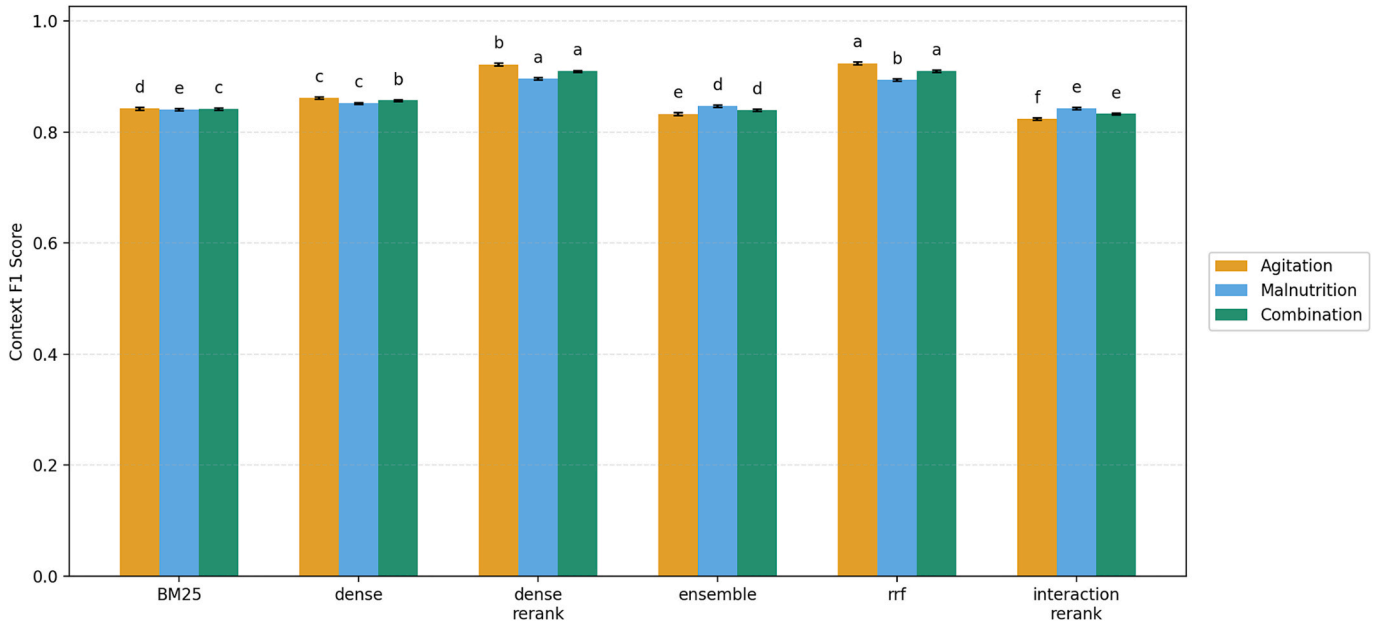


Fig. 2. Context F1 scores across retrieval methods for each task.

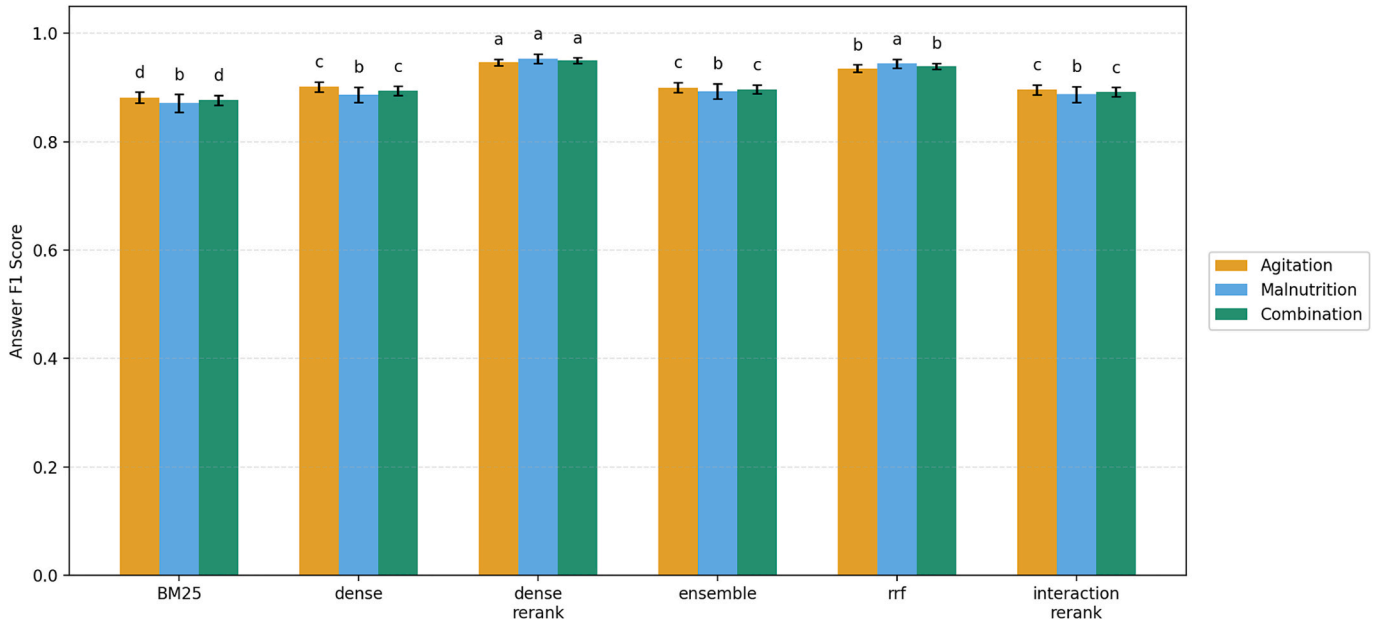


Fig. 3. Answer F1 scores across retrieval methods for each task.

that reranking not only improves relevance but also more faithfully grounds answers in retrieved evidence.

4.6. Contextual diversity

Contextual diversity measures the lexical diversity among retrieved contexts and also shows significant method differences ($p < 0.001$). For agitation (Fig. 5), dense-rerank and RRF achieved the most diverse contexts (0.895 and 0.886), while ensemble had the lowest diversity (0.761). In the malnutrition task, dense-rerank and RRF again produced the highest diversity (0.870 and 0.862), while ensemble, BM25, and interaction-rerank clustered at the bottom level (0.748–0.764). The pooled results from the two tasks further confirmed that dense-rerank was the top method (0.883), whereas ensemble had the lowest score (0.755). These findings may suggest that although diversity can broaden evidence coverage, it translates into downstream performance gains only when it is semantically aligned with the task need. This observation is consistent with prior findings that increased diversity alone does not guarantee improved task performance unless it remains aligned with task-relevant semantics [55,56].

4.7. Item-level accuracy

Finally, significant differences in item-level accuracy were observed across retrieval strategies ($p < 0.001$). For agitation (Fig. 6), dense-rerank, RRF, dense, and interaction-rerank clustered together at the top (0.964, 0.950, 0.908, 0.906), while ensemble and BM25 were at the lower end (0.865 and 0.840). For malnutrition, RRF and dense-rerank achieved the highest scores (0.944 and 0.941), significantly outperforming other methods (0.835–0.866). In the pooled analysis of the

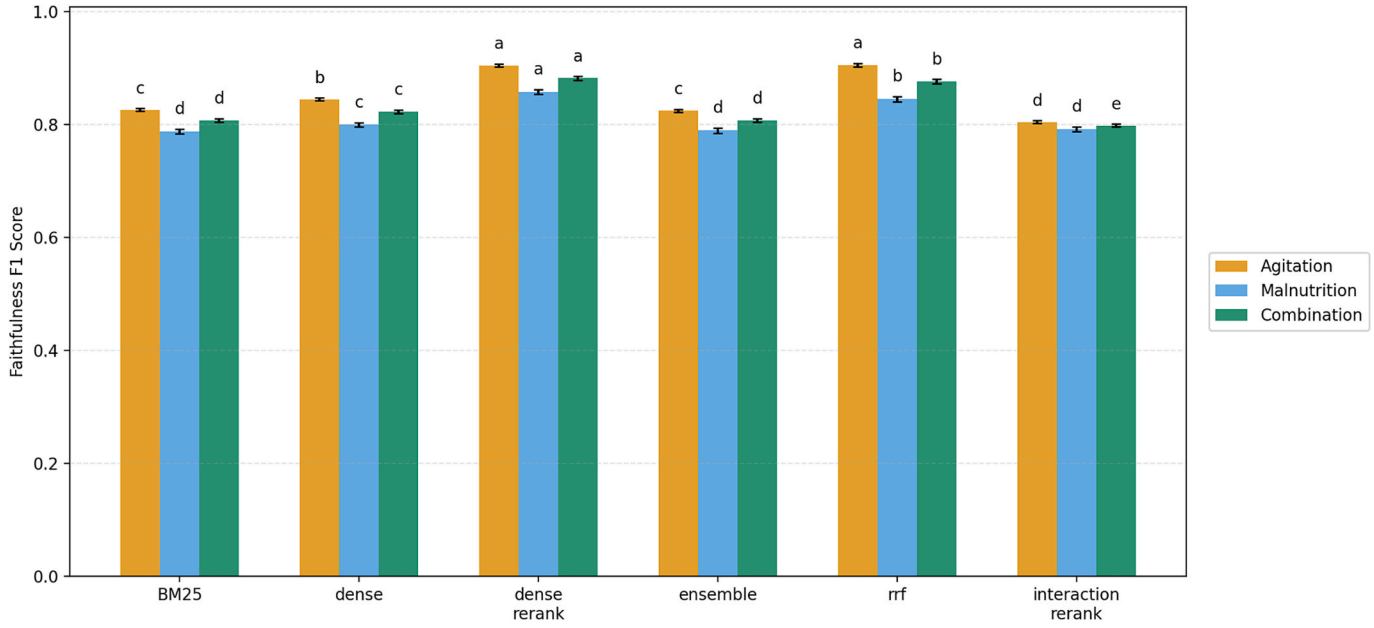


Fig. 4. Faithfulness F1 scores across retrieval methods for each task.

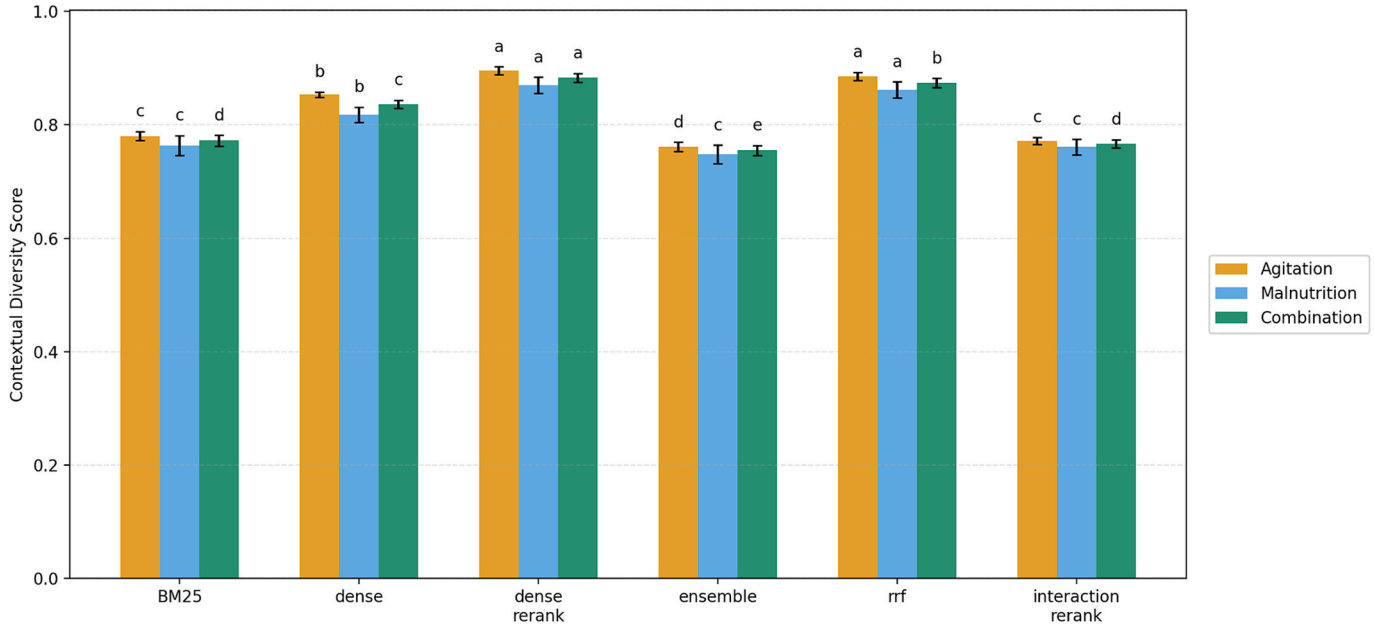


Fig. 5. Contextual Diversity scores across retrieval methods for each task.

two clinical tasks, dense-rerank and RRF again formed the top group (0.953 and 0.947), while BM25 was the weakest baseline (0.838). These results reinforce the robustness of reranking-based methods across tasks and when data are combined.

Retrieval methods achieved highly significant differences across all evaluation metrics. Dense-rerank consistently outperformed alternatives, with RRF at the top, while BM25 and ensemble were the weakest baselines.

The pooled analysis demonstrates that these patterns generalise across both agitation and malnutrition tasks, underscoring the superiority of reranking-based retrieval for clinical IE (see Table 1).

To further quantify the extent to which the retrieval module contributes to downstream extraction quality, beyond differences among retrieval strategies, we additionally evaluated a prompting-only baseline that performs extraction without retrieval augmentation.

4.8. Error analysis

We conducted a manual review of the erroneous outputs generated in the clinical information extraction (IE) tasks (see Supplementary Table 2). From this analysis, we identified three types of model errors: intrinsic fabrications, extrinsic fabrications, and false negatives. Intrinsic fabrications occur when the model misrepresents information that is present in the source text, e.g., distorting a finding, inverting a relationship, or incorrectly synthesising details from different parts of the input [57]. Extrinsic fabrications, by contrast, occur when the model introduces information without a basis in the source material, effectively inventing content [57,58]. False negatives arise when the model omits essential or clinically relevant information that is clearly stated in the original text [57], such as failing to capture a diagnosis, symptom, or treatment detail when summarising a clinical note.

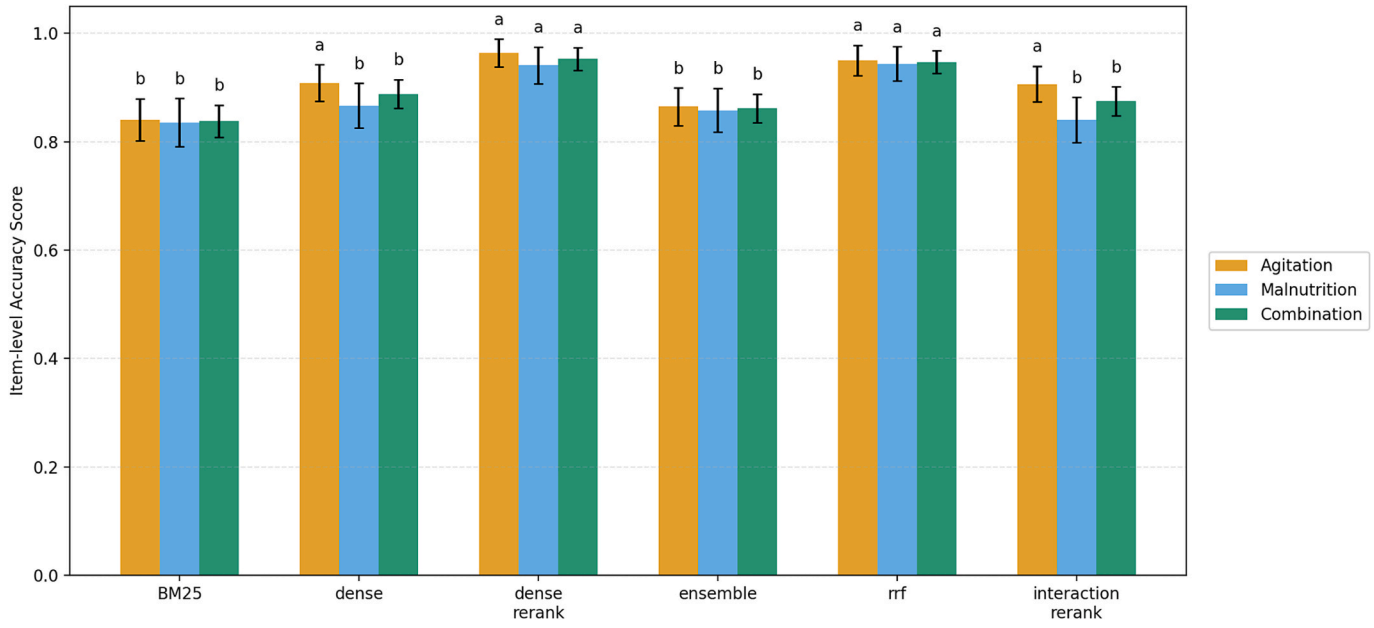


Fig. 6. Item-level Accuracy scores across retrieval methods for each task.

Table 1
Summarisation of the performance of each retrieval method for each clinical IE task (extraction of agitation symptoms in dementia and malnutrition risk factors).

Retrieval	Tasks	Context F1	Answer F1	Faithfulness F1	Contextual Diversity F1	Item-Level Accuracy
BM25	Agiation	0.842	0.881	0.826	0.78	0.84
	Malnutrition	0.841	0.871	0.787	0.764	0.835
Dense Retrieval	Agiation	0.861	0.901	0.844	0.853	0.908
	Malnutrition	0.852	0.886	0.8	0.818	0.866
Dense Retrieval + Rerank	Agiation	**0.922**	**0.946**	**0.904**	**0.895**	**0.963**
	Malnutrition	**0.896**	**0.953**	**0.858**	0.87	0.941
Dynamic Ensemble	Agiation	0.832	0.9	0.824	0.761	0.865
	Malnutrition	0.847	0.893	0.789	0.748	0.858
RRF	Agiation	0.924	0.935	0.905	**0.886**	0.95
	Malnutrition	0.894	0.944	0.845	0.862	**0.944**
Hybrid BM25 + Dense Retrieval	Agiation	0.824	0.896	0.804	0.772	0.906
	Malnutrition	0.843	0.887	0.792	0.761	0.84

As summarised in Table 2, intrinsic fabrications were the most prevalent error type across all retrieval methods, followed by extrinsic fabrications, whereas false negatives occurred less frequently.

Table 2
Types of model errors.

RAG Methods	Intrinsic Fabrications	Extrinsic Fabrications	False Negative
Sparse Retrieval (BM25)	67%	6%	27%
Dense Retrieval Using Semantic Embeddings	76%	16%	8%
Dense Retrieval with Cross-encoder Reranking (Dense Reranking)	50%	33%	17%
Reciprocal Rank Fusion (RRF)	66%	25%	9%
Hybrid BM25 to Dense Bi-encoder Rerank	79%	14%	7%

5. Discussion

Research from the same team on the same clinical tasks, using the same dataset, has demonstrated that the RAG approach with few-shot prompt learning achieved the same level of performance as fine-tuning an LLM with zero-shot learning [5,6]. This study further evaluated optimal retrieval strategies for clinical named-entity recognition tasks. While multi-stage retrieval strategies incorporating reranking have been shown to outperform single-stage retrieval methods in general information retrieval settings [59], their behaviour within LLM-driven clinical information extraction pipelines in aged care settings remains poorly understood. This gap is particularly evident when such pipelines are applied to noisy, longitudinal clinical narratives and normalisation-oriented extraction tasks; tasks that require not only identifying relevant mentions in free text but also mapping those mentions to canonical concepts in a controlled vocabulary or ontology. Rather than proposing a novel retrieval architecture, this study contextualises and characterises the effects of six established retrieval strategies within information-extraction workflows for aged-care clinical notes.

The primary contribution lies in examining how these strategies interact with schema constraints and heterogeneous clinical documentation to influence error structure and performance stability. In this context, the heterogeneity analysis serves to situate retrieval performance within the known complexity of clinical narratives, rather than to introduce new diversity metrics [60]. In addition, the comparison with a general-domain reference corpus further confirmed that the aged-care clinical narratives analysed in this study exhibit substantially higher linguistic heterogeneity than normal text. This empirical observation reinforces the need to evaluate retrieval strategies within RAG pipelines, as heterogeneous clinical narratives require mechanisms to identify relevant evidence across highly variable linguistic expressions. This study systematically compares the performance of six retrieval strategies within an RAG-enhanced LLM framework for clinical information extraction. Our experiment was conducted on two domain-specific tasks: the extraction of agitation symptoms in dementia and the identification of malnutrition risk factors. By adopting an instance-level repeated-measures ANOVA with Holm-adjusted post-hoc tests, we provide a more statistically rigorous analysis than prior work that relied on task-level averages.

5.1. Methodological insights

Our study demonstrates that the retrieval method has a substantial impact on the performance of RAG-enhanced LLMs applied to clinical notes in the Australian aged care setting. The results consistently showed that reranking-based methods, especially dense rerank, significantly outperformed both sparse retrieval (BM25) and standard dense encoders. Reciprocal Rank Fusion also performed strongly, achieving the same rank as dense rerank in Context F1 and Faithfulness F1, whereas ensemble methods underperformed in Contextual Diversity. Among the six methods, interaction rerank performed weakest on Context F1 and Faithfulness F1. These findings not only confirm that advanced reranking architectures are more effective at extracting contextually aligned and faithful evidence than baseline approaches, but also align with recent evidence that dense rerankers deliver superior retrieval precision on domain-specific tasks [6,37].

However, the contribution of this study extends beyond confirming a known ranking of retrieval architectures. In general-domain IR benchmarks, reranking primarily improves the relevance of a single retrieved answer to a factoid query; the downstream consequence of a retrieval error is typically a wrong answer [26]. Clinical IE, by contrast, imposes two additional demands that fundamentally alter how retrieval quality propagates through the pipeline. First, clinical IE tasks are *normalisation-oriented*: the model must not only locate relevant mentions but also map them to canonical concepts in a controlled schema (e.g., mapping "won't eat" to *poor appetite* in the malnutrition ontology). A retrieval error that surfaces a thematically related but schema-irrelevant passage can trigger intrinsic fabrication — the model rephrases or generalises source content to fill a schema slot, producing a plausible but incorrect normalisation.

Second, clinical IE tasks are *aggregative*: a single nursing note may contain multiple extractable entities, requiring the retrieval system to surface passages covering diverse aspects of the note rather than converging on a single best answer. These two properties mean that retrieval errors in clinical IE compound differently than in open-domain question answering: they affect not only answer correctness but also the completeness and internal consistency of the structured output. The error analysis presented below provides direct evidence that these compounding effects are the primary mechanism driving the amplified performance gap between reranking-based and single-stage methods observed in this study.

Similar performance improvements with RAG-enhanced LLMs have also been observed in other clinical IE domains. For example, Zhan et al. introduced the RAMIE framework for extracting information about dietary supplements. They demonstrated that combining multi-task learning with RAG substantially improved F1 scores across several IE subtasks compared with fine-tuned LLMs alone [61]. Comparable advances have been reported with entity-oriented retrieval strategies. For example, Lopez et al. developed the CLEAR framework to augment retrieval with clinical entities. CLEAR framework improves extraction accuracy (average F1 $\approx$ 0.90) and reduces token usage and inference time by more than 70% compared with an embedding-based RAG [62]. Meanwhile, our work focuses on reranking to optimise contextual precision and faithfulness. These studies collectively demonstrate that tailored retrieval strategies, such as reranking, multi-task augmentation, or entity grounding, can improve LLMs' accuracy and efficiency in clinical IE.

5.2. Error Structure in Clinical Information Extraction versus General-domain Retrieval

In the clinical IE setting examined here, intrinsic fabrications were the most prevalent error type, as found in other settings [42]. This pattern aligns with the imperfect representation learning and erroneous decoding mechanisms identified by Ji et al. as contributors to intrinsic hallucination: encoders can learn wrong correlations between different parts of the input, and decoders can attend to the wrong part of the source, generating outputs that mix facts between two similar entities [42]. It is also consistent with the normalisation demands of clinical IE: when the extraction schema requires a canonical term and the source text uses an informal or ambiguous expression, the model is more likely to mis-normalise (intrinsic fabrication) than to invent an entity from nothing (extrinsic fabrication).

Extrinsic fabrications, though less frequent, arise from three mechanisms that are well-documented in hallucination literature: (1) parametric knowledge bias, retrieval-generation conflict, when the retrieved document contradicts the model's parametric knowledge, causing the model to struggle to reconcile the two, sometimes defaulting to its internal and potentially outdated knowledge [63]; (2) long-tail knowledge gaps: LLMs perform well on common clinical concepts but are prone to fabrication when encountering rare or facility-specific entities (e.g., non-standard wound grading scales) that are underrepresented in training data; and (3) schema-driven over-completion: in the normalization-oriented extraction tasks examined here, the extraction schema itself can inadvertently elicit extrinsic fabrications [64] when the source text lacks information for a required field (e.g., a specific agitation term), prompting the model to generate a plausible value rather than returning null.

False negatives likely reflect the model's tendency to prioritise conciseness or salient patterns over completeness, missing less visible but clinically important details [65].

Dense rerank RAG directly addresses these error modes by constraining the generation space to retrieved evidence rather than parametric knowledge. The cross-encoder reranker acts as a precision filter, ensuring that passages provided to the model are tightly aligned with the extraction query and thereby reducing the ambiguity that drives both intrinsic misrepresentation and extrinsic confabulation.

Critically, this precision-filtering mechanism is more valuable in clinical IE than in general-domain QA precisely because clinical IE demands normalisation: a passage that is thematically relevant but terminologically imprecise can trigger mis-normalisation, whereas in open-domain QA the same passage might still yield a correct answer. Lewis et al. demonstrated a related effect when querying the United Kingdom's NICE clinical guidelines: reranking RAG achieved a faithfulness score of 0.95, an increase of 0.647 over non-RAG counterparts [66]. Our findings extend this observation by showing that the faithfulness advantage of reranking is amplified in normalisation-oriented clinical IE, where the cost of imprecise retrieval is not merely a less relevant answer but a structurally incorrect extraction.

5.3. Task-specific observations

The two clinical IE tasks, symptoms of agitation in dementia and malnutrition risk factors, differ in their linguistic and contextual characteristics, yet both exhibit similar performance patterns across retrieval methods. For agitation symptoms, dense rerank and RRF consistently yielded the highest context and answer F1 scores, while BM25 and interaction rerank were the weakest. For malnutrition, the relative advantage of reranking was equally pronounced, confirming the robustness of reranking-based retrieval within the context of aged-care clinical notes, despite differences in linguistic and contextual characteristics across the two evaluated information extraction tasks. Related findings were reported by Alkhalaf et al., who evaluated RAG for malnutrition management in RACFs using the same data set as this study and showed that, while RAG substantially improved summarisation accuracy, its benefit for extracting risk factors was more limited [6].

Although query complexity and specificity may influence absolute retrieval scores, the relative performance ordering of retrieval strategies remained stable across both extraction tasks. This aligns with our observation that although reranking enhanced overall accuracy, the relative gains in malnutrition risk factor extraction were less pronounced than for agitation symptoms. These findings should be interpreted in the context of schema-driven clinical IE queries, rather than as evidence of retrieval performance across arbitrary or free-form clinical question types. Across both tasks, the 208 test instances span multiple residents, time periods, and documentation styles, providing a heterogeneous but task-consistent evaluation set.

Notably, analysis of contextual diversity revealed that although ensemble retrieval produced the broadest lexical coverage, it did not translate into downstream performance gains. This finding echoes the earlier observations that diversity must remain semantically aligned to be beneficial [7]. The pooled analysis of two clinical tasks further enforces the superiority of reranking-based methods across tasks. Beyond overall performance gains, reranking-based methods exhibited more stable behaviour across tasks with differing extraction granularity, suggesting that cross-encoder refinement may be particularly effective at mitigating semantic drift introduced by heterogeneous clinical note structures.

5.4. Limitations

We acknowledge the following study limitations. First, this study evaluated only two clinical IE tasks: agitation symptoms in dementia (83 entities) and malnutrition risk factors (37 entities). Both are normalisation-oriented named entity recognition tasks operating on aged care nursing notes. While the results were largely consistent across these two tasks, they differ in ways that partially address concerns about task dependence: the agitation schema involves substantially greater terminological variability and a larger entity set than the malnutrition schema, and the two tasks produce queries of differing complexity and specificity. The stability of retrieval strategy rankings across these contrasting conditions provides some evidence of robustness. Nevertheless, two tasks are insufficient to support broadly generalisable conclusions about retrieval strategy performance across the full range of clinical IE scenarios.

In particular, this study did not evaluate tasks involving relation extraction, temporal reasoning, or open-ended clinical question answering, each of which would impose different demands on the retrieval component in terms of query structure, required context length, and the degree of cross-passage reasoning. Furthermore, the queries in both tasks are schema-driven, i.e., they are structured around a predefined entity list rather than formulated as free-form clinical questions, and retrieval performance is known to depend heavily on query complexity, specificity, and structure. The findings should therefore be interpreted as evidence of retrieval strategy behaviour within schema-driven clinical NER, rather than as generalisable claims about retrieval performance across arbitrary clinical question types. Future work should extend the evaluation to a broader range of clinical IE tasks with systematically varied query characteristics, including tasks with differing entity densities, longer reasoning chains, and free-form query formulations, to establish whether the observed advantages of reranking generalise beyond the normalisation-oriented setting examined here. Across both tasks in the present study, the 208 test instances span multiple residents, time periods, and documentation styles, providing a heterogeneous but task-consistent evaluation set.

Second, the hyperparameters in our study were fixed ($k = 3$, chunk size $= 250$). Without sensitivity analysis or ablation studies, the generalisability of the research conclusions regarding their influence on downstream performance is limited. Future work should explicitly evaluate the impact of various hyperparameter settings and adaptive strategies for balancing retrieval breadth and reranking precision. Additionally, Lu has shown that large language models struggle with token-level clinical named entity recognition, highlighting the broader challenge of applying LLMs to fine-grained extraction tasks in clinical text [67]. This further underscores the need to combine retrieval optimisation with domain-adapted modelling to improve entity-level reliability.

Third, the three F1-based evaluation metrics (Context, Answer, and Faithfulness) are all derived from the same BERTScore model and are therefore inherently correlated. As a result, they should not be interpreted as fully independent evaluation dimensions, and this dependence may limit the robustness of performance comparisons. Furthermore, the reliance on a single embedding-based automatic evaluation framework may introduce model-specific biases and operate without constrained decoding or human-in-the-loop assessment. As discussed in Section 3, constrained decoding was not applied because the lexical and structural heterogeneity of aged-care clinical notes violates the regularity assumptions on which such methods depend; however, its absence means that output consistency relies entirely on prompt-based schema specification, which may permit occasional format deviations that a constrained decoder would prevent. This may limit the interpretability and consistency of structured outputs, underscoring the importance of incorporating human-centred and model-diverse evaluation strategies in future work.

Finally, the comparison between clinical notes and normal text was conducted solely to characterise corpus-level heterogeneity, rather than for a comparable named entity recognition task. While this analysis provides useful contextual evidence that aged-care clinical notes are linguistically more heterogeneous than general-domain text, it does not directly evaluate how heterogeneity affects extraction performance in non-clinical corpora. Future studies could extend this comparison by applying comparable information extraction tasks across different text domains.

5.5. Implications

Unlike open-domain information retrieval benchmarks, which typically support a single factoid query, clinical information extraction tasks require context for both normalisation and aggregation. Retrieval errors thus propagate beyond answer correctness into downstream structuring and reasoning steps. This compounding effect offers a plausible explanation for the significant performance gap observed between reranking-based and single-stage retrieval methods in this study, and underscores the importance of retrieval strategy on clinical information extraction.

Against this backdrop, the findings of this research highlight the importance of rigorously selecting the retrieval method for LLM-based clinical IE pipelines operating on aged-care clinical notes. In particular, the optimal performance of dense reranking strategies suggests that cross-encoder rerank refinement is a practical and effective means of enhancing contextual grounding and factual faithfulness when extracting structured information from the two evaluated clinical IE tasks. Because this improvement resides in the retrieval component rather

than the downstream LLM architecture, practitioners and system designers working with similar clinical documentation can adopt dense reranking without redesigning their generative pipeline.

6. Conclusions

This study systematically evaluated the impact of six retrieval strategies on the performance of LLM-based retrieval-augmented generation (RAG) systems for clinical named entity recognition in the Australian aged care setting. Focusing on two high-impact use cases, extracting symptoms of agitation in dementia and malnutrition risk factors, we assessed the quality, faithfulness, and contextual diversity of extracted information from six retrieval methods and manually analysed the error causes.

The experiments revealed statistically significant differences among retrieval methods, with dense reranking consistently achieving the best performance across metrics. Reciprocal rank fusion (RRF) is frequently ranked among the top performers. At the same time, BM25 and ensemble approaches were consistently weaker. These results highlight that retrieval design is not merely a technical choice but a decisive factor in shaping downstream clinical information extraction. Our findings suggest that reranking-based retrieval frameworks were the optimal choice for the clinical named-entity recognition task in the Australian aged care setting. Beyond benchmarking, this work provides a reusable evaluation pipeline that integrates instance-level statistical testing to enable transparent, robust comparisons across methods. Our findings offer practical guidance for deploying retrieval strategies in real-world RAG applications, particularly in aged care and other low-resource clinical domains. They also provide a methodological foundation for downstream predictive modelling tasks, such as forecasting agitation onset in older people living in residential aged care facilities, based on extracted symptom and risk-factor trajectories from free-text clinical notes.

Statement of significance

Aspect	Description
Problem or Issue	Clinical notes contain rich but unstructured information that is difficult to extract automatically, especially for complex conditions such as agitation in dementia and malnutrition risk factors in aged care.
What is Already Known	Large language models (LLMs) can improve information extraction but are limited by unreliable retrieval; few studies have compared different retrieval strategies in real-world clinical note contexts.
What this Paper Adds	This study systematically compares six retrieval-augmented generation (RAG) strategies, sparse, dense, dense reranking, and three hybrid methods, across two aged-care information extraction tasks, identifying dense reranking as the most effective and reliable approach.
Who will Benefit from this Knowledge	Biomedical informatics researchers, clinical AI developers, and healthcare practitioners can use these insights to design more accurate and trustworthy RAG pipelines for clinical documentation analysis and predictive modelling.

CRediT authorship contribution statement

Hengyi Zhang: Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Dinithi Vithanage:** Writing – original draft, Methodology, Investigation, Conceptualization. **Chao Deng:** Writing – review & editing, Supervision, Conceptualization. **Ping Yu:** Writing – review & editing, Validation, Supervision, Software, Resources, Project administration, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The data that has been used is confidential.

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